

Matukumilli V. D. Prasad

Materials Theory Group
Theoretical Sciences Unit
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EMPLOYMENT:

- | | |
|--|----------------------------|
| Research Associate
Theoretical Sciences Unit,
Jawaharlal Nehru Centre for Advanced Scientific Research (JNCASR), Bengaluru. | May 2017 — Present |
| Research Consultant
Advanced Technology Development Centre,
Indian Institute of Technology Kharagpur. | Aug 2015 — May 2017 |
| Junior Engineer
SYNERGIES CASTINGS Limited,
VSEZ, Visakhapatnam, AP. | May 2006 — May 2007 |

RESEARCH INTEREST:

- Nanoscale friction [[Nano Lett. 16 \(2016\) 2174](#)] [[Nano Lett. 17 \(2017\) 2131](#)]
- Thermal transport [[JACS 147 \(2025\) 3758](#)] [[JACS 141 \(2019\) 20293](#)] [[Angew. Chem. 61 \(2022\) e202200071](#)]
- Quantum geometry

EDUCATION:

- | | |
|---|--------------------|
| Indian Institute of Technology Kharagpur, WB <ul style="list-style-type: none">• Ph.D., Thesis: "Phonon Scattering Dynamics of Sliding Motion in Carbon Nanotube Oscillators"• Advisor: Prof. Baidurya Bhattacharya.
Thesis Committee Members: Prof. L S Ramachandra, Prof. Nirupam Chakraborti, Prof. Sanjoy Bandyopadhyay | 2008 — 2015 |
| Andhra University College of Engineering, Visakhapatnam, AP <ul style="list-style-type: none">• B.E. in Mechanical Engineering. | 2001 — 2005 |

GRANTS EXPERIENCE:

- **Research Consultant** (ISRO SPONSORED PROJECT, **PI: Baidurya Bhattacharya**; Budget INR 20L, 04/2015 — 03/2017): "Development of Constitutive Equations for CNT Epoxy Nanocomposites"
- **Jr. Project Officer** (DMRL SPONSORED PROJECT, **PI: Baidurya Bhattacharya**; Budget INR 10L, 11/2008 — 06/2009): "Simulation Studies of Mechanical Behavior and Failure of Carbon Nanotubes"

- Debattam Sarkar, Subarna Das, Vaishali Taneja, Manisha Samanta, Koushik Jagadish, Animesh Das, Monika Bhakar, **Matukumilli V. D. Prasad**, Suresh Perumal, Goutam Sheet, Dirtha Sanyal, Koushik Pal, N Ravishankar, Umesh V. Waghmare, Kanishka Biswas, "Glassy Thermal Transport Triggers Ultra-High Thermoelectric Performance in GeTe", *Advanced Materials* **37**, 12 (2025)
- Animesh Bhui, **Matukumilli V. D. Prasad**, Shuva Biswas, Abdul Ahad, Anupama Ghata, Divya Rawat, Moinak Dutta, Ajay Soni, Umesh V. Waghmare, Kanishka Biswas, "s-d Coupling-Induced Dynamic Off-Centering of Cu Drives High Thermoelectric Performance in TiCuS", *Journal of the American Chemical Society* **147**, 4 (2025)
- Moinak Dutta, **Matukumilli V. D. Prasad**, Juhi Pandey, Ajay Soni, Umesh V. Waghmare, and Kanishka Biswas, "Local Symmetry Breaking Suppresses Thermal Conductivity in Crystalline Solids", *Angewandte Chemie*. **134**, 15 (2022)
- Moinak Dutta, Shidaling Mattheppanavar, **Matukumilli V. D. Prasad**, Juhi Pandey, Avinash Warankar, Pankaj Mandal, Ajay Soni, Umesh V. Waghmare, and Kanishka Biswas, "Ultralow Thermal Conductivity in Chain-like TiSe Due to Inherent Ti⁺ Rattling", *Journal of the American Chemical Society* **141**, 51 (2019)
- **Matukumilli V. D. Prasad*** and Baidurya Bhattacharya, "Phononic Origins of Friction in Carbon Nanotube Oscillators", *Nano Lett.* **17**, 2131 (2017)
– Media hits: featured story by [Nanowerk](#), [Nanotechweb](#), [IITKGP News](#)
- **Matukumilli V. D. Prasad*** and Baidurya Bhattacharya, "Phonon Scattering Dynamics of Thermophoretic Motion in Carbon Nanotube Oscillators", *Nano Lett.* **16**, 2174 (2016)
- **Matukumilli V. D. Prasad** and Baidurya Bhattacharya, "Phonon Wave-Packet Scattering and Energy Dissipation Dynamics in Carbon Nanotube Oscillators", *J. Appl. Phys.* **118**, 244906 (2015)
- **Matukumilli V. D. Prasad** and Baidurya Bhattacharya, "Buckling and Post-Buckling Characteristics of Cantilevered Single-walled Carbon Nanotubes in Bending", *Int. J. Nanosci.* **14**, 1550019, (2015)
- **Matukumilli V. D. Prasad** and Baidurya Bhattacharya, "Molecular Dynamics Simulations of Carbon Nanotube-based Oscillators having Topological Defects", *Int. J. Nanosci.* **10**, 355 (2011)

INVITED BOOK CHAPTERS

- **Matukumilli V. D. Prasad** and Umesh V. Waghmare, "Theory and simulations of lattice thermal conduction". In: ed. by Paolo Mele, Dario Narducci, Michihiro Ohta, Kanishka Biswas, Juan Morante, Shrikant Saini, Tamio Endo, *Thermoelectric thin films: materials and devices* Springer, Jun 2019.

AWARDS AND HONORS:

- **Outstanding Research Award** 2016-17 (National PARAM Supercomputing Facility, Pune)
– *one among the top five outstanding research works carried-out using PARAM YUVA II in the year 2016-17*
- **Best Poster Presentation Award** Nov 2016 (National Symposium on Ultrasonics, S. N. Bose National Centre for Basic Sciences, Kolkata)

TEACHING EXPERIENCE:

Teaching assistant

December 2016.

- GIAN course on Nonequilibrium Statistical Mechanics and Molecular Dynamics, organized by IIT Kharagpur, Winter 2016

Student Mentorship, IIT Kharagpur

2013 — 14.

- Undergraduate student: Nishanth K S. I worked with Nishanth on a problem investigating thermal transport characteristics of metal nanoclusters encapsulated inside carbon nanotube using molecular dynamics.

PROFESSIONAL ACTIVITIES AND SERVICE:

- Reviewer for *ACS Applied Energy Materials*, *Smart Materials and Structures*, *Nanotechnology*, *Materials Research Express*, *Computational Materials Science* and *Nanoscale*.
- Student Organizing Committee, MACB-2020, JNCASR (Feb 2020): Celebrated the 80th birthday of Prof. Mike Klein
- Member of American Physical Society (APS), Indian Society for Heat and Mass Transfer (ISHMT), American Society of Mechanical Engineers (ASME)

INVITED TALKS AND SEMINARS: (* SPEAKER)

- **Matukumilli V. D. Prasad**, and Baidurya Bhattacharya*. "*Mechanical Properties of CNTs and CNT Nanocomposites Through ab-initio and Atomic Simulations*", Dept of Civil Engg, Johns Hopkins University, Baltimore, MD, USA, Oct 2012
- **Matukumilli V. D. Prasad**, and Baidurya Bhattacharya*. "*Molecular Dynamics Simulations of Thermal Transport in Carbon Nanotubes*", Annual Technical Meeting of NMD-ATM, College of Engineering Pune, Pune, India, Nov 2014
- **Matukumilli V. D. Prasad**, and Baidurya Bhattacharya*. "*Phonon Scattering Dynamics of Friction and Thermophoretic Motion in Carbon Nanotube Oscillators*", International Centre for Mathematical Sciences, Edinburgh, UK, Oct 2015
- **Matukumilli V. D. Prasad***. "*Carbon Nanotube Oscillators: Phonons and Friction*", Journal Club Seminar at Physics Department, IIT Kharagpur, India, Oct 2016

- **Matukumilli V. D. Prasad***, and Baidurya Bhattacharya. *"Molecular dynamics of thermal-phonon induced drive: Coherent and diffusive scattering regimes in nanomechanical devices"*, 4th International Conference on Advances in Materials & Materials Processing, IIT Kharagpur, India, Nov 2016
- **Matukumilli V. D. Prasad***, and Umesh V. Waghmare. *"Phonon drag effect for thermophoretic and thermoelectric energy conversion applications"*, Frontiers of Mechanics colloquium, IIT Kharagpur, India, 24-26 Nov 2017
- **Matukumilli V. D. Prasad**, and Umesh V. Waghmare*. *"Anharmonic phonon thermal transport"*, India-UK workshop on Thermoelectric Materials for Waste-Heat Harvesting, JNCASR, India, Jan 2018
- **Matukumilli V. D. Prasad***, and Baidurya Bhattacharya, *"Development of Constitutive Equations for CNT-Epoxy Nanocomposites"*, Project Closure Presentation, Vikram Sarabhai Space Center (VSSC), Thiruvananthapuram, India, Feb 2018

CONFERENCE ORAL/POSTER PRESENTATIONS: (* SPEAKER)

- **Matukumilli V. D. Prasad***, and Baidurya Bhattacharya. *"Molecular Dynamics Simulations of Ni Nanowire Bending"*, Molecules and Materials – New Directions, Jawaharlal Nehru Centre for Advanced Scientific Research, Bengaluru, India, Dec 2008
- **Matukumilli V. D. Prasad***, and Baidurya Bhattacharya. *"Bending Properties of Carbon Nanotubes with Multiple Vacancy Defects using Atomistic Simulations"*, AIMTDR 2008, 2nd International 23rd All India Manufacturing Technology, Design and Research Conference, IIT Madras, India, Dec 2008
- **Matukumilli V. D. Prasad***, and Baidurya Bhattacharya. *"Molecular Dynamics Simulations of Carbon Nanotube-based Oscillators having Topological Defects"*, ICANN 2011, International Conference on Advanced Nanomaterials and Nanotechnology, IIT Guwahati, India, Nov 2011
- **Matukumilli V. D. Prasad***, and Baidurya Bhattacharya. *"Energy Dissipation and Phononic Friction in Carbon Nanotube Oscillators"*, ICTACEM 2014, 6th International Conference on Theoretical, Applied, Computational and Experimental Mechanics, IIT Kharagpur, India, Dec 2014
- **Matukumilli V. D. Prasad***, and Baidurya Bhattacharya. *"Carbon Nanotube Oscillators: Effect of Small Bending Strain"*, ICTACEM 2014, 6th International Conference on Theoretical, Applied, Computational and Experimental Mechanics, IIT Kharagpur, India, Dec 2014
- **Matukumilli V. D. Prasad***, and Baidurya Bhattacharya. *"Acoustic Phonon Propagation and Interfacial Scattering in Nanostructures"*, NSU 2016, National Symposium on Ultrasonics, S. N. Bose National Centre for Basic Sciences, Kolkata, India, Nov 2016 **Awarded best Poster Presentation/Prize**
- **Matukumilli V. D. Prasad**. Attended the QPC 2017, Conference on Quantum Physics and Consciousness, Bhaktivedanta Institute & Bharatiya Vidya Mandir, Kolkata, India, Feb 2017
- **Matukumilli V. D. Prasad***, Koushik Pal, and Umesh V. Waghmare. *"Phonon thermal transport in ternary Zintl compounds: Bi_2Mg_2X ($X = Mg, Ca$)"*, India-UK workshop on Thermoelectric Materials for Waste-Heat Harvesting, JNCASR, India, Jan 2018

WORKSHOPS/SCHOOLS:

- “Mechanics over Micro and Nano Scales (STCMMN 09)”, Bengal Engineering and Science University, Shibpur, Kolkata, Dec 2009
- “Electronic Structure and Properties of Low Dimensional systems”, Physics Department, IIT Kharagpur, Apr 2016
- “Lithium Ion Battery Technology and Mathematical Modeling”, Metallurgical and Materials Engineering Department, IIT Kharagpur, May 2016
- “Winter School-2017 on Frontiers in Materials Science”, JNCASR, Dec 2017
- “India-UK workshop on Thermoelectric Materials for Waste-Heat Harvesting”, New Chemistry Unit, JNCASR, Jan 2018
- “Sorbonne-JNCASR School on Advanced Computational Materials Science (SJSACMS-2018)”, Thematic Unit of Excellence in Computational Materials Science, JNCASR, Feb 2018
- “Young Scientist Discussion Meeting on Chemistry of Nanomaterials”, New Chemistry Unit, JNCASR, Jul 2018
- Winter School on “Modelling and Simulations of Materials for Energy and Environment”, Theoretical Sciences Unit, JNCASR, Dec 2018
- “Ten years of ICMS celebrations and Annual SAMat symposium”, International Center for Materials Science, JNCASR, Jan 2019
- “Modern Approaches in Chemistry & Biology-2020 (MACB-2020)” as a celebration of the 80th birthday of Professor Mike Klein, JNCASR, Feb 2020
- “Moving ions with VASP” Online VASP workshop, September 27-30, 2022
- “New Approaches and Machine learning Methods for Ab initio calculations: NAMMA Psi-k Workshop 2023” IISc and JNCASR, July 2023
- “MD@60” JNCASR-CECAM Conference, February 2024
- “International School on Modelling and Simulations of Materials” JNCASR, January 2025

COMPUTATIONAL SKILLS:

- **First-Principles and Atomistic Simulations:** Extensive experience with VASP, QUANTUM ESPRESSO, and classical molecular dynamics packages including LAMMPS, GROMACS, GULP, and MEDEA; complemented by self-developed MD and MC codes for system-specific simulations.
- **Phonon and Thermal Transport Analysis:** Proficient in PHONOPY, PHONO3PY, D3Q, THERMAL2, PYTHON-SSCHA, PHONTS, SHENGBTE, ALMABTE, TDEP and ALAMODE.
- **Programming & Scripting:** Fortran, C, C++, and MPI parallel programming; experienced in Python-based workflows for high-throughput and data analysis.
- **Mathematical and Data Analysis Tools:** MATLAB/Octave, Mathematica, GNUplot, Matplotlib, and xmgrace.
- **Structure and Trajectory Visualization:** ATOMEYE, VMD, V-SIM, JMOL, XCRYSDEN, OVITO, VESTA, and ATOMSK.
- **Operating Systems:** Advanced user of Linux-based environments; experienced with Windows.